Bundled Service Structures and Price Variation from CMS Machine-Readable Files: A North Carolina Case Study

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***Abstract*—The CMS Hospital Price Transparency Rule requires hospitals to post machine-readable files (MRFs) with standard charges for all services. However, most analyses treat individual codes in isolation, while real encounters like colonoscopies comprise bundled services with facility, professional, anesthesia, and pathology components. This paper presents a methodology for reconstructing encounter-level service groupings from MRFs using revenue codes and regex-based crosswalks, and for quantifying component-level price variation. Analyzing five North Carolina hospitals, we find substantial between-hospital variation, within-hospital negotiated price gaps across insurers (log ratios up to 6.13), and incomplete component representation. Results demonstrate that bundle-based analysis—rather than single-code comparison—is essential for meaningful price transparency and consumer decision-making.**

***Index Terms*—Machine-Readable Files, Healthcare, Chargemaster, CMS, Pricing Transparency.**

I. Introduction

The 2021 CMS Hospital Price Transparency Rule requires hospitals to post machine-readable files (MRFs) with gross charges, cash prices, and payer-specific negotiated rates [1]. However, most analyses treat individual CPT/HCPCS codes in isolation, while real encounters (colonoscopy, MRI, CT) comprise multiple bundled components: facility fees, professional services, anesthesia, and pathology [2]. This disconnect between line-item prices and true encounter costs obscures variation and limits transparency.

Prior work documents substantial price variation: Linde et al. report coefficients of variation exceeding 100% [4], while Wang et al. find within-hospital negotiated price ratios exceeding 3:1 across insurers [7]. However, few studies recover bundle structure directly from MRFs or visualize component-level heterogeneity.

We analyze five North Carolina hospitals to: (1) develop a data-driven methodology for reconstructing bundles from revenue-code signals and regex-based crosswalks; (2) decompose price variation across hospitals, payers, and components using statistical testing; and (3) identify implications for consumer tools and reference pricing. Only the top 100-200 Lab codes, CPT codes, and HCPCS codes billed for 2024 were analyzed.

II. Related Work

MRFs lack explicit bundle encoding [3], complicating consumer interpretation. Prior studies document extreme cross-hospital variation (CV $>$ 100%) [4], [5] and within-hospital insurer gaps (max/min ratios $>$ 3:1) [7]. Colonoscopy bundles include facility, professional, anesthesia, and pathology fees [6], [9], [10], but most analyses rely on claims data rather than recovering bundles from MRFs. This study aims to infer bundles from revenue codes (RC), regex addendum cross-walks, and decompose variation using statistical testing and visualization.

III. Methods

A. Service Grouping via Revenue Codes and Regex Crosswalks

Because MRFs do not explicitly encode encounter-level bundles, we reconstructed clinically coherent service groupings by combining revenue-code signals and description-based regex crosswalks. For each targeted shoppable service, we selected its "index" Lab/CPT/HCPCS code (e.g., 45378 or 45380 for colonoscopy) and, within each hospital, identified associated lines that plausibly belong to the same encounter.

1) Normalize code systems and price fields and standardize payer–plan identifiers.
2) Identify candidate facility versus professional lines using revenue codes and hospital-defined service groupings; map common families (e.g., facility/technical, professional, anesthesia, pathology).
3) Apply regex-based crosswalks to descriptions and published addenda (e.g., CMS Addendum B) to unify synonyms and local nomenclature; link add-on and ancillary codes to the index service.
4) Merge candidates into a service bundle when they share the index code stem, a consistent revenue-code family, or a description match indicating the same encounter context within the same hospital.

The output of this procedure associates each Lab/CPT/HCPCS line with an inferred component type derived from revenue-code families and description matches.

These mappings were hand-validated against standard colonoscopy and imaging episode definitions from ASPE [9]. Sensitivity analyses varied regex strictness and revenue-code inclusion lists; results were stable to reasonable perturbations.

B. Bundle-Level Pricing and Component Signatures

For each hospital and service, we aggregated negotiated prices by component type to construct a "bundle signature." Specifically, for each component c in hospital h, we computed:

$$\text{MedianPrice}_{h,c} = \text{median}\left\{\text{negotiated prices of codes in component } c\right\}.$$

For each bundle, we computed distributional signatures (mean, median, standard deviation, interquartile range, and coefficient of variation) separately for each component type and price type.

C. Price Variation Metrics

To evaluate cross-hospital and within-hospital variation, we computed:

1) Coefficient of Variation (CV): Defined as:

$$\text{CV} = \frac{\sigma}{\mu},$$

following Linde et al. [4]. CV was computed for each component × hospital × price-type combination.

2) Within-Hospital Negotiated Price Gaps: For each hospital and service, we computed:

$$\text{Gap}_h = \log\left(\frac{\max_{\text{payer}}(\text{negotiated price})}{\min_{\text{payer}}(\text{negotiated price})}\right).$$

These metrics align with those used in Wang et al. [7].

3) Hospital Deviation Scores: For each hospital h and component c, we computed deviation from the statewide median:

$$\Delta_{h,c} = \log\left(\frac{\text{MedianPrice}_{h,c}}{\text{MedianPrice}_{\text{NC median},c}}\right).$$

D. Statistical Testing

To evaluate whether differences in bundle or component prices across hospitals were statistically significant, we applied:

1) Kruskal–Wallis H-test: Non-parametric test for comparing distributions across multiple hospitals:

$$H = \frac{12}{N(N+1)}\sum_{i=1}^{k} n_i(R_i - \bar{R})^2,$$

where R_i is the mean rank for hospital i. We applied false-discovery-rate (FDR) correction across all component types using Benjamini–Hochberg.

2) Levene's Test (Variance Equality): Used optionally to assess cross-hospital variance differences, particularly for facility and anesthesia fees.

3) Log-Ratio One-Sample Tests: We tested whether log(price ratios) differed significantly from 0, indicating systematic markups or discounts.

IV. Results

It should be noted that not all hospitals offer the measured billable services as described by CMS.gov, thus not all tables will contain statistic information for them, see Table I. Additionally, some hospitals purposefully obfuscate their MRF data—requiring a mapping addendum—which can lead to loss of information without the proper key-code pair, or an available CPT code document on hand. This vastly limits the ability to concatenate the data and standardize it.

TABLE I
COMPREHENSIVE

Category	Count
Records with Missing Data	15529
Total Records in Dataset	97836
Percent Records with Missing Data	15.87%
Missing Estimated Amount	15457
Missing Max Price	73
Missing Min Price	73
Missing All Three Prices	1

A. Bundle Structures

Using the procedure described in the methods, we defined distinct bundle structures for each targeted shoppable service. For screening and diagnostic colonoscopy, bundles typically contained between three and six components, corresponding to facility fees, professional services, anesthesia, pathology, and ancillary services such as biopsy processing. Imaging bundles for MRI and CT procedures were somewhat simpler, often combining a dominant facility component with one or more professional codes.

Table II summarizes the number of recovered bundles per service and the distribution of component counts across hospitals. Most services yielded a single dominant bundle family per hospital, but in some cases we observed alternative structures (e.g., separate bundles with and without anesthesia codes), reflecting local practice patterns or coding conventions.

TABLE II
SUMMARY OF RECOVERED BUNDLES BY SERVICE AND HOSPITAL.

Service	Bundles	Med. Comp.	IQR
Endoscopy – Colonoscopy	4	1.0	0.0
Imaging – MRI	4	1.0	0.0
Imaging – CT	4	1.0	0.0
Metabolic Panel	5	1.0	0.0
COVID-19 PCR	4	1.0	0.0
Lipid Panel	4	1.0	0.0

B. Component-Level Price Variation Across Hospitals

We next examined component-level price variation across hospitals using the coefficient of variation (CV) and interquartile ranges. For each component, we aggregated payer-specific negotiated prices within hospital and then summarized these component medians across hospitals.

Overall, we observed substantial dispersion in component prices for all studied services. Facility fees for colonoscopy and CT showed particularly wide variability, with high between-hospital dispersion relative to professional and pathology components.

Table III reports example summary statistics by component, including mean and median negotiated prices, IQR, and CV.

TABLE III
COMPONENT-LEVEL NEGOTIATED PRICE VARIATION.

Component	Mean	Median	IQR	CV
CPT	385	130	263	3.67
HCPCS	2.1K	99	619	4.21
RC	17.5K	316	23.6K	0.63

Note: Prices in USD. K=thousands

C. Statistical Significance of Cross-Hospital Differences

To assess whether observed differences across hospitals were statistically significant, we applied the Kruskal–Wallis test to component-level negotiated prices for each bundle, with false-discovery-rate correction across tests.

Table V presents an example of the Kruskal–Wallis H statistics, unadjusted p-values, and FDR-adjusted q-values for a subset of bundles and components. For most core components (especially facility fees), cross-hospital differences in negotiated prices remained highly significant after FDR correction, indicating that the observed dispersion is unlikely to be driven by sampling noise alone.

D. Hospital-Level Deviation Patterns

Using hospital deviation scores $\Delta_{h,c}$, we compared each hospital's component prices to the statewide median for that component. Table IV summarizes deviation scores by hospital, reporting median deviation across components and the range of deviations observed. Some hospitals consistently priced above the statewide median for facility components while remaining closer to the median for professional services; others displayed the opposite pattern or more heterogeneous profiles. Note that Wake Med. North did not have estimated costs for any plan, thus deviation could not be calculated.

TABLE IV
HOSPITAL DEVIATION SCORES VS. NC MEDIAN.

Hospital	Med. Δ	Range
AdventHealth	-0.46	10.58
Duke Univ.	-0.53	0.83
Wake Med. North	NA	NA
UNC Med. Ctr.	0.39	1.09
UNC Rex	0.33	1.66

E. Within-Hospital Payer-Plan Heterogeneity

Table VI reports summary statistics for these within-hospital gaps. Many hospital–bundle combinations exhibit substantial spread between the least and most expensive payer plans, indicating that insurer identity remains a key determinant of patient liability for identical services.

Heatmaps of the median, maximum, and minimum payer-plan prices reveal distinct clusters of high-priced and low-priced plans (Figure 1, Figure 2, Figure 3). These patterns echo findings from Wang et al. [7] and demonstrate that meaningful heterogeneity exists even after controlling for hospital and bundle. Rather than clustering tightly into a small number of homogeneous price groups, plans display dense blocks of both positive and negative log ratios, suggesting that hospitals negotiate idiosyncratic combinations of high and low rates across clinical categories. This pattern is inconsistent with simple "high-paying vs. low-paying payer" narratives and instead points toward a multidimensional contracting space in which relative advantage depends on the specific composition of laboratory, diagnostic, pharmaceutical, and ancillary components within a bundle.

V. DISCUSSION

The Kruskal–Wallis tests reported in Table V reveal statistically significant differences in negotiated prices across hospitals for nearly all tested bundles after FDR correction. For high-volume categories such as injectable drugs (J-codes), DNA/NAAT testing, and flow cytometry, the test statistics exceed several hundred, with p-values effectively at machine precision. These findings align with prior empirical work documenting pervasive cross-hospital dispersion in standard charges and negotiated rates [4], [5]. However, our results extend this literature by demonstrating that such variation persists even when services are aggregated into clinically coherent bundles rather than analyzed as isolated CPT or HCPCS codes.

Perhaps the most striking finding of this analysis is the magnitude of within-hospital negotiated price variation across insurers and plans. Table VI documents log gaps ranging from 1.01 to 6.13 for common laboratory and diagnostic bundles. A log gap of 6.13 (AdventHealth, COVID-19 PCR) corresponds to a ratio of approximately $e^{6.13} \approx 460$, implying that the highest-paying plan reimburses nearly 460 times more than the lowest-paying plan for the same service at the same facility. Even for more typical cases, median log gaps of 2–3 indicate price ratios on the order of 7–20. One-sample tests confirm systematic dispersion (t ≈ 16.05, p $\approx 2.22 \times 10^{-40}$), echoing Wang et al. [7]. This heterogeneity exceeds between-hospital variation, complicating consumer tools that display single average rates. Effective transparency requires payer-plan-specific displays personalized to user coverage.

HCPCS codes show highest relative dispersion (CV = 4.21), while RC median is \$316 despite mean $>$ \$17.5K (Table III) likely due to revenue bundles being inconsistent in capturing total cost of care. Many bundles showed simple structures (median 1.0 component), suggesting incomplete representation of anesthesia or pathology services that would appear in real claims. This gap limits MRF-based transparency, reducing realistic cost comparisons on-top of missing cost estimates seen in Table I. The min-ratio matrix (Figure 3) supports this distinction by exhibiting sharper, polarized banding. The concentration of extreme values in contiguous horizontal and vertical stripes indicates that certain payers consistently secure the lowest reimbursements for at least some codes regardless of comparator, while others exhibit the opposite pattern.

Bundle-based displays should aggregate all encounter components into total price estimates, following CMS shoppable service guidance [2]. Reference pricing should use bundle totals stratified by plan type, not single-code charges. Cash prices sometimes exceeded negotiated rates, offering little benefit to self-pay patients.

A. Limitations

Several limitations must be acknowledged. First, this analysis is based on a convenience sample of five North Carolina hospitals and may not generalize to other states or market structures. North Carolina's hospital market includes large academic medical centers, regional health systems, and independent community hospitals, but it lacks certain characteristics common in other states, such as dominant for-profit chains or highly consolidated insurance markets. Replicating this analysis with broader geographic coverage would strengthen external validity.

Second, our ETL pipeline required substantial manual harmonization of payer and plan identifiers. Hospitals use inconsistent naming conventions (e.g., "BCBS NC Blue Value", "Blue Cross Blue Shield NC Value Plan", "BCBSNC Value") that complicate aggregation. While we applied string similarity and domain knowledge to construct a crosswalk, some misclassification is inevitable. Standardized payer and plan identifiers—such as the use of Health Plan Identifiers (HPID) or Taxpayer Identification Numbers (TIN)—would greatly improve the reliability of cross-hospital comparisons.

Third, the revenue-code and regex-based grouping approach is inherently limited by the information present in MRFs. Hospitals do not report which lines are linked on actual claims; instead, we infer grouping from shared revenue codes, description patterns, or hospital-defined service groupings. This approach works well for tightly defined service categories (e.g., laboratory panels) but may fail to capture loosely coupled components (e.g., separate professional and anesthesia billing). Access to claims-level line-linkage data would enable more robust bundle definitions.

Fourth, we focus exclusively on negotiated prices and do not account for utilization, quality, or patient outcomes. A low-priced hospital may appear attractive from a cost perspective but may have higher complication rates or lower patient satisfaction. Integrating MRF-based price transparency with quality metrics (e.g., Hospital Compare, Leapfrog scores) remains an important direction for future tool development.

Finally, our analysis aggregates payer-specific negotiated rates but does not distinguish between in-network and out-of-network pricing, nor does it capture patient cost-sharing rules (deductibles, copays, coinsurance). The "negotiated rate" disclosed in an MRF is the amount the insurer pays the hospital, but the patient's out-of-pocket liability depends on benefit design. Consumer-facing transparency tools must integrate MRF data with plan-specific cost-sharing structures to generate personalized price estimates.

B. Conclusion

This study demonstrates that bundle-based analysis of MRFs reveals systematic price variation across hospitals, insurers, and service components that single-code comparisons obscure. Within-hospital payer heterogeneity (up to 460× ratios) exceeds between-hospital variation, requiring personalized, payer-specific transparency tools. Future work should link MRFs to claims data for validation, integrate quality metrics, develop ML-based bundle construction methods, and test whether bundle-focused displays improve consumer decision-making and reduce surprise billing.

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APPENDIX

TABLE V
KRUSKAL-WALLIS TEST RESULTS (173 SIGNIFICANT CODES, SHOWING TOP/BOTTOM 10).

Code	Type	Service	n_h	n	Mean	Med.	CV	H	p	RBC
87798	Lab	BK/JC Virus PCR	4	4524	226	160	0.93	1675	$<10^{-300}$	3
88185	Lab	FLOWCYTOMETRY/TC ADD-ON	4	2268	122	58	1.00	795	10^{-172}	36
82784	Lab	ASSAY IGA/IGD/IGG ...	4	929	96	92	0.76	559	10^{-121}	95
87150	Lab	DNA/RNA AMPLIFIED PROBE	4	1010	551	103	2.19	520	10^{-112}	35
87186	Lab	Yeast Suscept.	4	1108	75	23	1.03	498	10^{-107}	83
Q9966	HCPCS	Contrast Iodixanol	4	925	240	160	1.34	439	10^{-95}	141
82570	Lab	Creatinine	4	517	58	16	1.02	402	10^{-87}	67
83735	Lab	Magnesium	4	746	103	37	1.42	365	10^{-79}	84
Q9967	HCPCS	Iodixanol 320mg	4	2905	264	38	2.23	362	10^{-78}	6
J7512	HCPCS	Prednisone 1mg	4	879	115	4	2.36	324	10^{-70}	15
			...							
81443	Lab	GENETIC TESTING SEVERE ...	2	107	8.3K	6.4K	0.89	6	10^{-2}	35
88189	Lab	FLOWCYTOMETRY/READ 16	2	47	119	106	0.49	6	10^{-2}	88
J0174	HCPCS	Lecanemab	2	74	823	669	0.35	5	10^{-2}	25
J9030	HCPCS	BCG Live 50mg	4	160	2.5K	625	2.23	9	10^{-2}	126
0037U	Lab	TARGET GENETIC SEQ ...	2	87	8.1K	6.1K	0.68	5	10^{-2}	41
J9173	HCPCS	Durvalumab	3	102	3.7K	3.8K	0.56	7	10^{-2}	49
J1010	HCPCS	Methylpred. Acet.	4	873	158	27	1.81	9	10^{-2}	7
87500	Lab	Vanco Resist	2	86	450	188	1.82	4	10^{-2}	87
J7626	HCPCS	Budesonide	4	356	26	12	1.35	7	10^{-2}	33
J7508	HCPCS	Tacrolimus	2	57	29	26	0.95	3	10^{-2}	23

Type: Lab = Laboratory, HCPCS = Healthcare Common Procedure Coding System. n_h = number of hospitals. n = number of price observations.H = Kruskal-Wallis statistic. p = unadjusted p-value. RBC $\rightarrow$ *Rank by Charges* = rank order by total charges across hospitals from the top 100-200 billed codes for 2024. Note that RBC is provided for context and is not used in the Kruskal-Wallis test, although it can be seen that the frequency at which a code is charged does not impact the price discrepancy.

TABLE VI
WITHIN-HOSPITAL PAYER-PLAN PRICE GAPS (LOG RATIOS, SHOWING TOP/BOTTOM 10).

Hospital	Code	Description	n_p	Min ($)	Med. ($)	Max ($)	Ratio	Log Gap
AdventHealth	J7517	MYCOPHENOLATE MOFETIL ...	18	0	15	258.6K	2.58E5	17.07
AdventHealth	Q0138	FERUMOXYTOL 510 MG/17M...	18	0	184	128.2K	1.28E5	16.37
AdventHealth	Q5110	FILGRASTIM-AAFI 300 MC...	18	0	128	70.5K	7.05E4	15.77
AdventHealth	29827	HC SHLDR ARTHROSCOP,SU...	18	0	10.8K	31.4K	3.14E4	14.96
Duke Univ. Hosp.	81443	Mutation Analysis Atax...	12	0	2.6K	26.8K	2.68E4	14.80
AdventHealth	J9030	BCG LIVE 50 MG IS SUSR	18	0	42	23.2K	2.32E4	14.66
AdventHealth	J0878	DAPTOMYCIN 350 MG IV SOLR	18	0	27	21.6K	2.16E4	14.59
AdventHealth	33285	HC INSERTION SUBQ CARD...	18	0	9.6K	18.2K	1.82E4	14.41
AdventHealth	36902	HC INTRO CATH DIALYSIS...	18	0	8.6K	18.1K	1.81E4	14.41
AdventHealth	J2919	METHYLPREDNISOLONE SOD...	18	0	26	14.8K	1.48E4	14.21
			...					
Duke Univ. Hosp.	J1306	INCLISIRAN 284 MG/1.5 ...	12	7.9K	12.7K	21.8K	2.7	1.01
Duke Univ. Hosp.	J0174	LECANEMAB-IRMB 100 MG/...	12	607	966	1.7K	2.7	1.01
Duke Univ. Hosp.	88344	Immunoperoxidase per S...	12	240	393	657	2.7	1.01
Duke Univ. Hosp.	J2356	TEZEPELUMAB-EKKO 210 M...	12	9.6K	21.6K	26.4K	2.7	1.01
Duke Univ. Hosp.	88309	Surg Path Gross Micro ...	12	389	746	1.1K	2.7	1.01
Duke Univ. Hosp.	88377	Fish Quant/Semiquant E...	12	390	779	1.1K	2.7	1.01
Duke Univ. Hosp.	87150	Cult Type ID by Amp Probe	12	74	121	203	2.7	1.01
UNC Rex Hosp.	J9119	CEMIPLIMAB-RWLC 50 MG/...	11	9.4K	9.9K	25.7K	2.7	1.01
AdventHealth	J9271	HC INJ PEMBROLIZUMAB	18	48	52	131	2.7	1.00
UNC Rex Hosp.	J9144	DARATUMUMAB 1,800 MG-H...	11	8.9K	9.4K	24.2K	2.7	1.00

Note: Prices in USD. K=thousands. n_p = number of payer-plans. Ratio = Max/Min. Log Gap = $\log(\text{Max}/\text{Min})$.

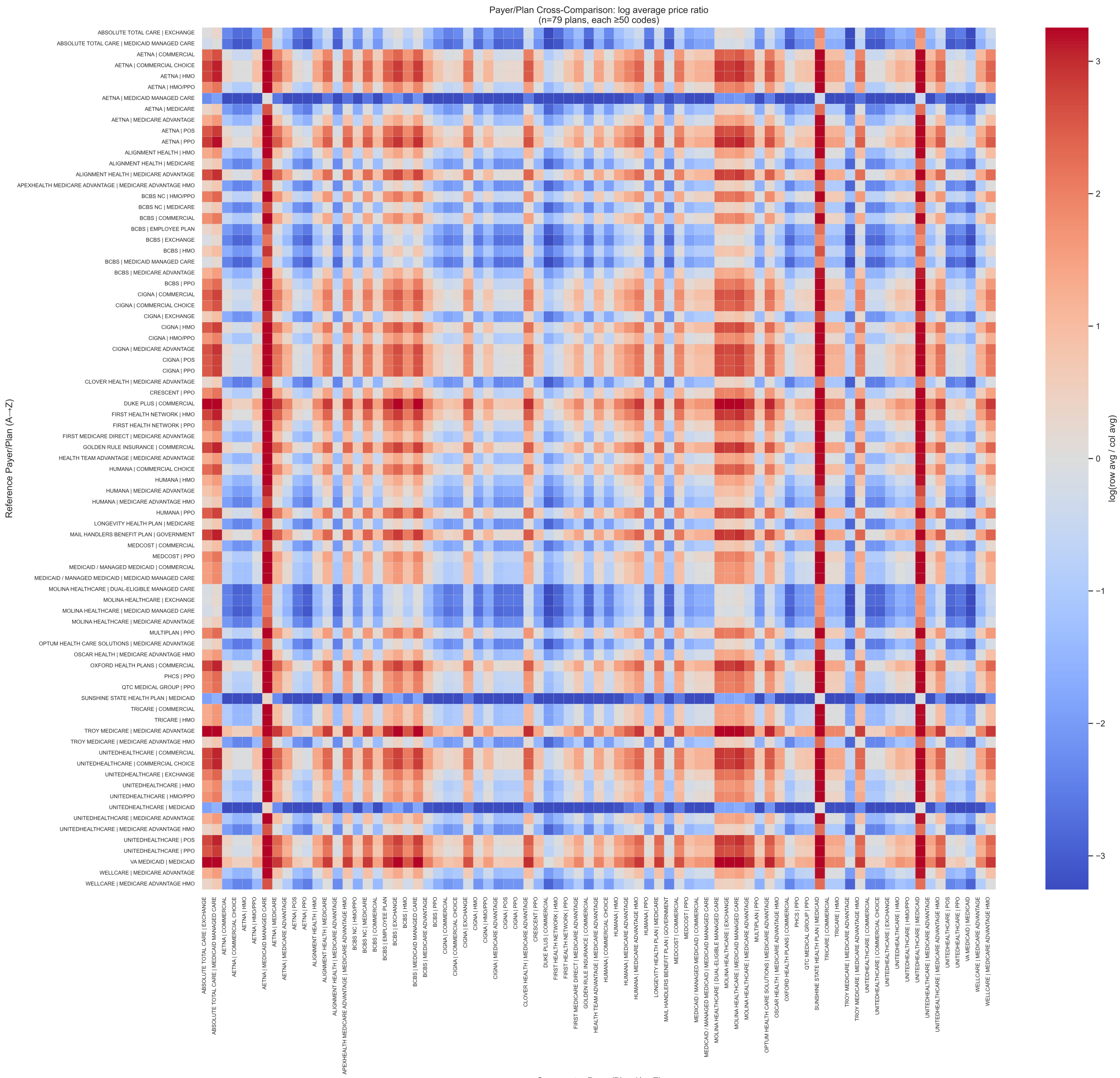

Fig. 1. Payer/Plan Cross-Correlation: log median price ratio. Median-based correlation matrix revealing the most stable patterns in payer-plan pricing relationships across hospitals and service categories. Red indicates the column feature is more expensive (log base) than the row feature, blue indicates it is less expensive (log base).

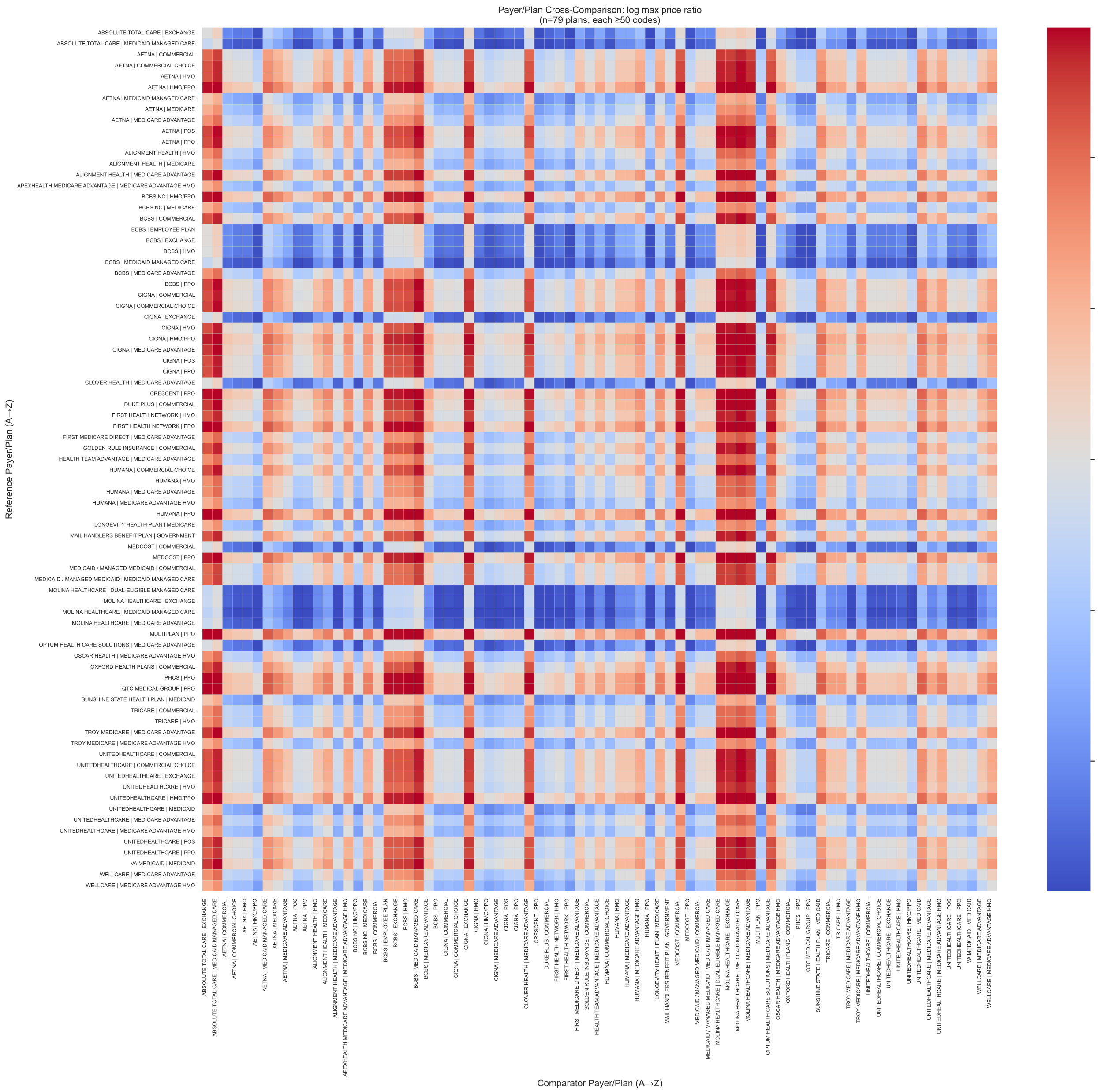

Fig. 2. Payer/Plan Cross-Correlation: log max price ratio. Heatmap displaying the correlation structure of maximum negotiated prices across payer-plans and service bundles, revealing systematic pricing patterns and outliers. Red indicates the column feature is more expensive (log base) than the row feature, blue indicates it is less expensive (log base).

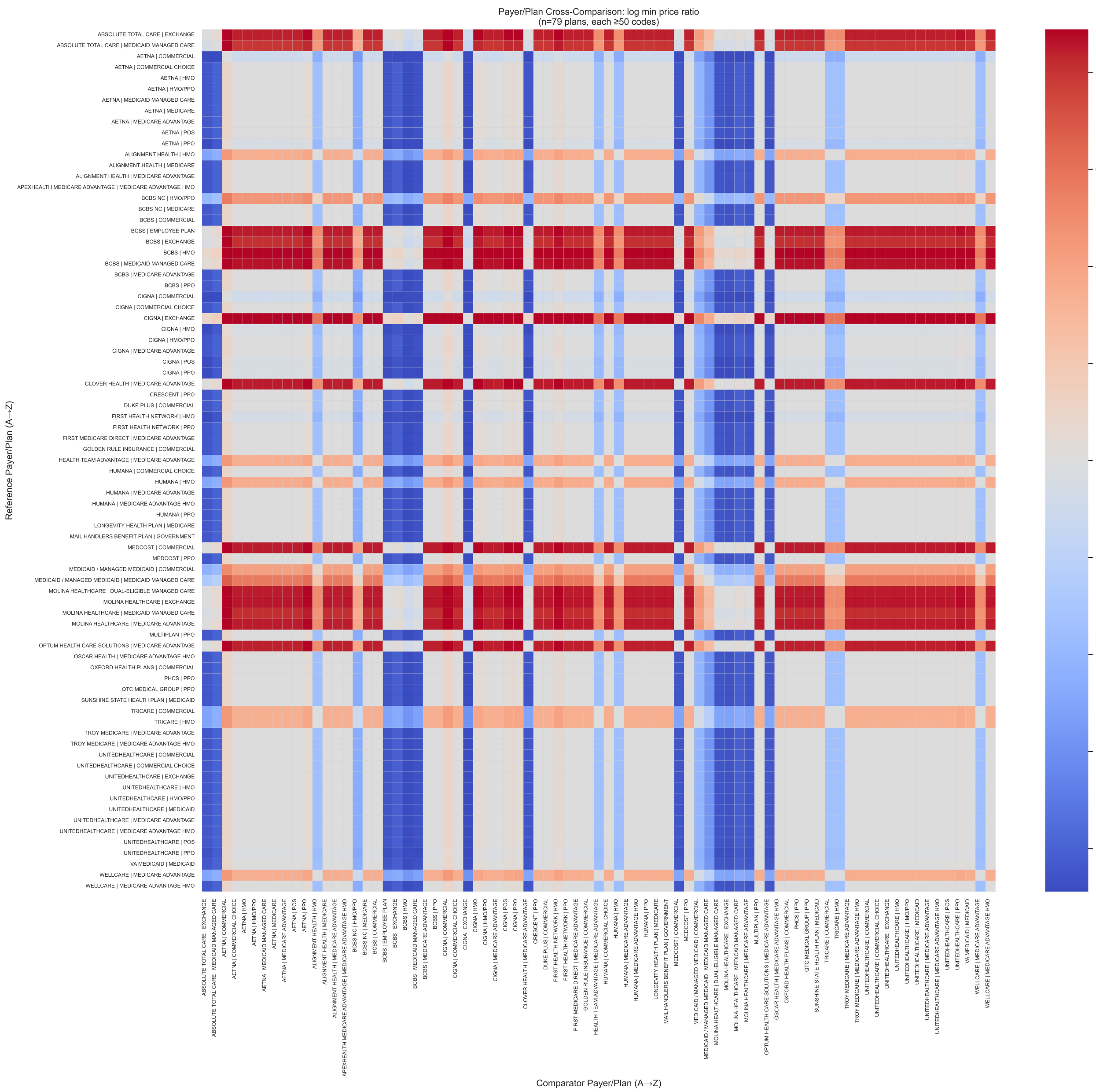

Fig. 3. Payer/Plan Cross-Correlation: log min price ratio. Heatmap showing the correlation structure of minimum negotiated prices across payer-plans and service bundles, revealing baseline pricing patterns. Red indicates the column feature is more expensive (log base) than the row feature, blue indicates it is less expensive (log base).